Selection Sort

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In Selection Sort, the selection process is to select the minimum position or maximum position element from i'th index to (N-1)th index. After selection we swap the element from the minimum position or maximum position with i'th position element. And this process will continue for (N-1) time. After completion of the process our Array is either sorted in either ascending order or in descending order.

For example: Ascending order

0	1	2	3	4	5	6	7	8
79	45	56	36	70	17	29	12	35

i th position → 0; minimum position → 7

0	1	2	3	4	5	6	7	8
79	45	56	36	70	17	29	12	35

8

0	1	2	3	4	5	6	7	8
12	45	56	36	70	17	29	79	35

7

so,

0	1	2	3	4	5	6	7	8
12	17	56	36	70	45	29	79	35

6

0	1	2	3	4	5	6	7	8
12	17	29	36	70	45	56	79	35

5

0	1	2	3	4	5	6	7	8
12	17	29	35	70	45	56	79	36

4

0	1	2	3	4	5	6	7	8
12	17	29	35	36	45	56	79	70

3

0	1	2	3	4	5	6	7	8
12	17	29	35	36	45	56	79	70

2

0	1	2	3	4	5	6	7	8
12	17	29	35	36	45	56	79	70

1

0	1	2	3	4	5	6	7	8
12	17	29	35	36	45	56	70	79

Total no. of comparison: $\boxed{\frac{N*(N-1)}{2}} = \frac{9 \times \cancel{8}\,4}{\cancel{2}} = 36$ steps

∴ Performance = $O(N^2)$

A

0	1	2	3	4	5	6	7	8
12	45	56	36	70	17	29	79	35

i = 0

minp = ~~0~~ ~~1~~ ~~2~~ ~~3~~ ~~4~~ ~~5~~ 7

j = ~~1~~ ~~2~~ ~~3~~ ~~4~~ ~~5~~ ~~6~~ ~~7~~ ~~8~~ 9

t = 79

Ascending order:

```
public static void selectionSort(int A[]){
        int i, j, t, minp;
        for(i = 0; i < A.length - 1; i++){
            for(minp = i, j = i+1; j < A.length; j++){
                if(A[minp] > A[j]){
                    minp = j;
                }
            }
            //we got the minp, so we swap the minp with i'th position
            t = A[i];
            A[i] = A[minp];
            A[minp] = t;
        }
    }
```

Descending order:

```
public static void selectionSort(int A[]){
        int i, j, t, maxp;
        for(i = 0; i < A.length - 1; i++){
            for(maxp = i, j = i+1; j < A.length; j++){
```

Do the dry run for this function using the given array:

0	1	2	3	4	5	6	7	8
79	45	56	36	70	17	29	12	35

```
                if(A[maxp] < A[j]){
                    maxp = j;
                }
            }
            //we got the minp, so we swap the minp with i'th position
            t = A[i];
            A[i] = A[maxp];
            A[maxp] = t;
        }
    }
```

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cending order:

Do the dry run for this function w the given array:

0	1	2	3	4	5	6	7	8
~~79~~	45	56	36	70	17	29	12	35

```
lic static void selectionSort(int A[]){
    int i, j, t, maxp;
    for(i = 0; i < A.length - 1; i++){
        for(maxp = i, j = i+1; j < A.length; j++){
```

i = 0
maxp = 0
j = ~~1~~ ~~2~~ ~~3~~ ~~4~~ ~~5~~ ~~6~~ ~~7~~ ~~8~~
t = 79

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```
            if(A[maxp] < A[j]){
                maxp = j;
            }
        }
        //we got the minp, so we swap the minp with i'th position
        t = A[i];
        A[i] = A[maxp];
        A[maxp] = t;
    }
}
```

Inserting and deleting an element from the array:

Inserting

	0	1	2	3	4	5	6	7	8	9	10
A	60	25	42								

insert 44 at index 1

	0	1	2	3	4	5	6	7	8	9	10
A	60	44	25	42							

public void insertElement (int A[], int indexAt, int ele){

}

→ insert 28 at 4

	0	1	2	3	4	5	6	7	8	9	10
A	60	44	25	42	28						

insert 15 at 0

	0	1	2	3	4	5	6	7	8	9	10
A	15	60	44	25	42	28					

insert 90 at 10 ← valid index

	0	1	2	3	4	5	6	7	8	9	10
A	15	60	44	25	42	28	90				

insert 90 at 100 ← "Not a valid index"

	0	1	2	3	4	5	6	7	8	9	10
A	15	60	44	25	42	28					90

insert 57 at -1 ← Not a valid index.

Rule of array is "size must be predefined".

this variable will keep the track how many elements are filled

A.length ← number of elements.

```
static int filled = -1;
public static void insertElement (int A[], int indexAt, int ele) {
        // Check for empty cells in the array before inserting elements
}
```

```
Enter size of array: 5

Press 1 to insert element at specified
position.
Press 2 to delete element at specified
position.
Press 3 to display array.
Press 0 to exit.
Enter your choice (0-3) : 1

Enter position to insert: 10
```

```
Enter element to insert: 45

Invalid position.
Press 1 to insert element at specified
position.
Press 2 to delete element at specified
position.
Press 3 to display array.
Press 0 to exit.
Enter your choice (0-3) : 1

Enter position to insert: 4

Enter element to insert: 100

Element inserted successfully at 0 position.
Press 1 to insert element at specified
position.
Press 2 to delete element at specified
position.
Press 3 to display array.
Press 0 to exit.
Enter your choice (0-3) : 3

Array A[]:
[100, 0, 0, 0, 0]

Press 1 to insert element at specified
position.
Press 2 to delete element at specified
position.
Press 3 to display array.
Press 0 to exit.
Enter your choice (0-3) : 1

Enter position to insert: 0

Enter element to insert: 80

Element inserted successfully at 0 position.
Press 1 to insert element at specified
position.
Press 2 to delete element at specified
position.
Press 3 to display array.
Press 0 to exit.
Enter your choice (0-3) : 3

Array A[]:
[80, 100, 0, 0, 0]

Press 1 to insert element at specified
position.
Press 2 to delete element at specified
position.
Press 3 to display array.
Press 0 to exit.
Enter your choice (0-3) : 1

Enter position to insert: 1
```

```
Enter element to insert: 9

Element inserted successfully at 1 position.
Press 1 to insert element at specified
position.
Press 2 to delete element at specified
position.
Press 3 to display array.
Press 0 to exit.
Enter your choice (0-3) : 3

Array A[]:
[80, 9, 100, 0, 0]

Press 1 to insert element at specified
position.
Press 2 to delete element at specified
position.
Press 3 to display array.
Press 0 to exit.
Enter your choice (0-3) : 1

Enter position to insert: 1

Enter element to insert: 190

Element inserted successfully at 1 position.
Press 1 to insert element at specified
position.
Press 2 to delete element at specified
position.
Press 3 to display array.
Press 0 to exit.
Enter your choice (0-3) : 3

Array A[]:
[80, 190, 9, 100, 0]

Press 1 to insert element at specified
position.
Press 2 to delete element at specified
position.
Press 3 to display array.
Press 0 to exit.
Enter your choice (0-3) : 1

Enter position to insert: 2

Enter element to insert: 300

Element inserted successfully at 2 position.
Press 1 to insert element at specified
position.
Press 2 to delete element at specified
position.
Press 3 to display array.
Press 0 to exit.
Enter your choice (0-3) : 3

Array A[]:
[80, 190, 300, 9, 100]
```

```
Press 1 to insert element at specified
position.
Press 2 to delete element at specified
position.
Press 3 to display array.
Press 0 to exit.
Enter your choice (0-3) : 1

Enter position to insert: 3

Enter element to insert: 400

Array is full!
Press 1 to insert element at specified
position.
Press 2 to delete element at specified
position.
Press 3 to display array.
Press 0 to exit.
Enter your choice (0-3) : 0
```